

ECON0100 | Homework B5 | Market Changes

| Solutions

Answers and work are shown in the red blocks.

Butterbeer

Preferences for butter beer can be represented by the following demand curve:

$$P = 100 - \frac{1}{2}Q_D$$

The marginal cost of butter beer can be represented by the following equation:

$$P = 20 + \frac{1}{2}Q_S$$

Prices are in galleons and quantity is in bottles.

| From the curves, $Q_d = 200 - 2P$ and $Q_s = 2P - 40$. Equilibrium is $Q^* = 80$ bottles at $P^* = 60$ galleons.

Q1 | Price Elasticity

a) What is the price elasticity of demand when the price changes from 10 to 30? _____

| $-1/4$. Q goes from 180 to 140: $\varepsilon = \frac{-40/160}{20/20} = -\frac{1}{4}$.

b) What is the price elasticity of demand when the price changes from 40 to 60? _____

| -1 . Q goes from 120 to 80: $\varepsilon = \frac{-40/100}{20/50} = -1$.

c) What is the price elasticity of demand when the price changes from 70 to 90? _____

| -4 . Q goes from 60 to 20: $\varepsilon = \frac{-40/40}{20/80} = -4$.

d) Is demand elastic, unit elastic, or inelastic in a), b), and c)?

a) inelastic, b) unit elastic, c) elastic. Each price change is 20 galleons and each quantity change is 40 bottles, the same slope throughout, yet demand grows more elastic higher up the line.

e) What is the price elasticity of supply when the price changes from 40 to 60? _____

! 5/3, elastic. Q goes from 40 to 80: $\varepsilon = \frac{40/60}{20/50} = \frac{5}{3}$.

Q2 | Supply & Demand Shifters

For each change below, which curve in the butter beer market shifts, and in which direction?

a) Wages rise across the wizarding world, and butter beer is a normal good.

demand shifts up · demand shifts down · supply shifts up · supply shifts down

! Demand shifts up: buyers of a normal good buy more at every price when income rises.

b) Wages rise across the wizarding world, and butter beer is an inferior good.

demand shifts up · demand shifts down · supply shifts up · supply shifts down

! Demand shifts down: buyers of an inferior good switch to something better when income rises.

c) The price of pumpkin juice, which wizards drink instead of butter beer, rises.

demand shifts up · demand shifts down · supply shifts up · supply shifts down

! Demand shifts up: pumpkin juice is a substitute, so some of its buyers switch to butter beer.

d) The price of the pumpkin pasties that wizards eat with their butter beer rises.

demand shifts up · demand shifts down · supply shifts up · supply shifts down

! Demand shifts down: pasties are a complement, so fewer pasties means less butter beer at every price.

e) The rent on brewing cellars in Hogsmeade rises.

demand shifts up · demand shifts down · supply shifts up · supply shifts down

! Supply shifts up: a costlier input raises the cost of every bottle.

f) A new brewing charm lets brewers make each batch in half the time.

demand shifts up · demand shifts down · supply shifts up · supply shifts down

■ Supply shifts down: better technology lowers the cost of every bottle.

g) The price of firewhisky, which brewers could make instead, rises.

demand shifts up · demand shifts down · supply shifts up · supply shifts down

■ Supply shifts up: brewing butter beer now gives up more, so its opportunity cost rises.

Q3 | Comparative Statics

Butter beer has culturally been associated with a good time. This all changed when a high profile wizard went viral slamming the taste. At the same time an ingredient of the drink, butter extract, became incredibly difficult for butter beer makers to acquire. Use a graph to discuss the effect these two events had on the butter beer market.

a) Due to the viral post, the demand curve: *shifted up* · *stayed the same* · *shifted down*

■ Shifted down: a change in tastes lowers what buyers will pay for every bottle.

b) Due to the butter extract shortage, the supply curve: *shifted up* · *stayed the same* · *shifted down*

■ Shifted up: a scarcer input raises the cost of every bottle.

c) Without using numbers, how did the two events together change the market?

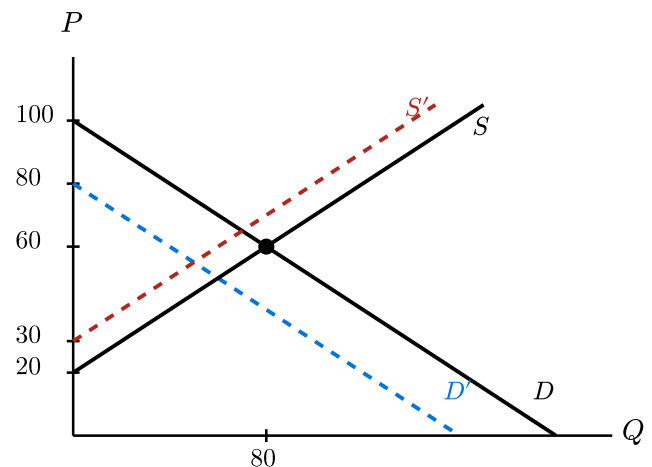
Prices: *increased* · *stayed the same* · *decreased* · *indeterminate*

Quantity: *increased* · *stayed the same* · *decreased* · *indeterminate*

Prices: indeterminate. Demand down pushes the price down, supply up pushes it up.

Quantity: decreased. Both shifts lower the quantity.

The graph shows both shifts: demand down to $P = 80 - \frac{1}{2}Q$, supply up to $P = 30 + \frac{1}{2}Q$.



d) Use numbers to compare the new equilibrium to the original one. After both events, the supply and demand curves became:

$$P = 30 + \frac{1}{2}Q_S \text{ and } P = 80 - \frac{1}{2}Q_D$$

What is the change in equilibrium price? _____

What is the change in equilibrium quantity? _____

Price: -5 galleons. Quantity: -30 bottles.

Originally $100 - \frac{1}{2}Q = 20 + \frac{1}{2}Q \rightarrow Q = 80$,
 $P = 60$. After both events $80 - \frac{1}{2}Q = 30 + \frac{1}{2}Q \rightarrow Q = 50$, $P = 55$.

With numbers the price question resolves: the viral post (down 20) outweighs the butter extract shortage (up 10), so the price ends below the original 60. The quantity falls, as c) said.

